

METABOLIC DYSREGULATION IN NEURODEGENERATION

- METAL DYSHOMEOSTASIS

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BACKGROUND

- Neurodegenerative disease impose a financial burden which is increasing with an aging global population. Thus, therapies that effectively target disruptions in energy metabolism, oxidative stress and cell death are vital [1,2].
- Manganese (Mn) and Zinc (Zn) are essential trace metals, but their dyshomeostasis contributes to neurotoxicity and manganese overload that have been shown to induce cytotoxicity and neurodegenerative disease [3,5].
- The metal transporter **SLC39A14 (ZIP14)** is crucial for Mn and Zn uptake and clearance, and mutations in humans have been shown to cause early-onset neurodegeneration [4,5].
- Despite decades of research into manganese toxicity, **precise molecular mechanisms** by which manganese and zinc dyshomeostasis disrupt [6] metabolism and **lead to neurodegenerative diseases** remain **poorly defined**.
- The **absence of clear molecular targets** remains a translational roadblock, driving high attrition rates in therapeutic pipelines [6].

RESEARCH QUESTION

How does a knockout of the SLC39A14 (ZIP14) transporter influence the metabolic profile of 4 dpf zebrafish embryos under manganese and zinc treatment?

AIMS

- Characterise metabolic changes caused by **manganese** treatment in the embryonic zebrafish (*Danio rerio*) ZIP14 KO model
- Characterise metabolic changes caused by **zinc** treatment in the embryonic zebrafish (*Danio rerio*) ZIP14 KO model

REFERENCES

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METHOD

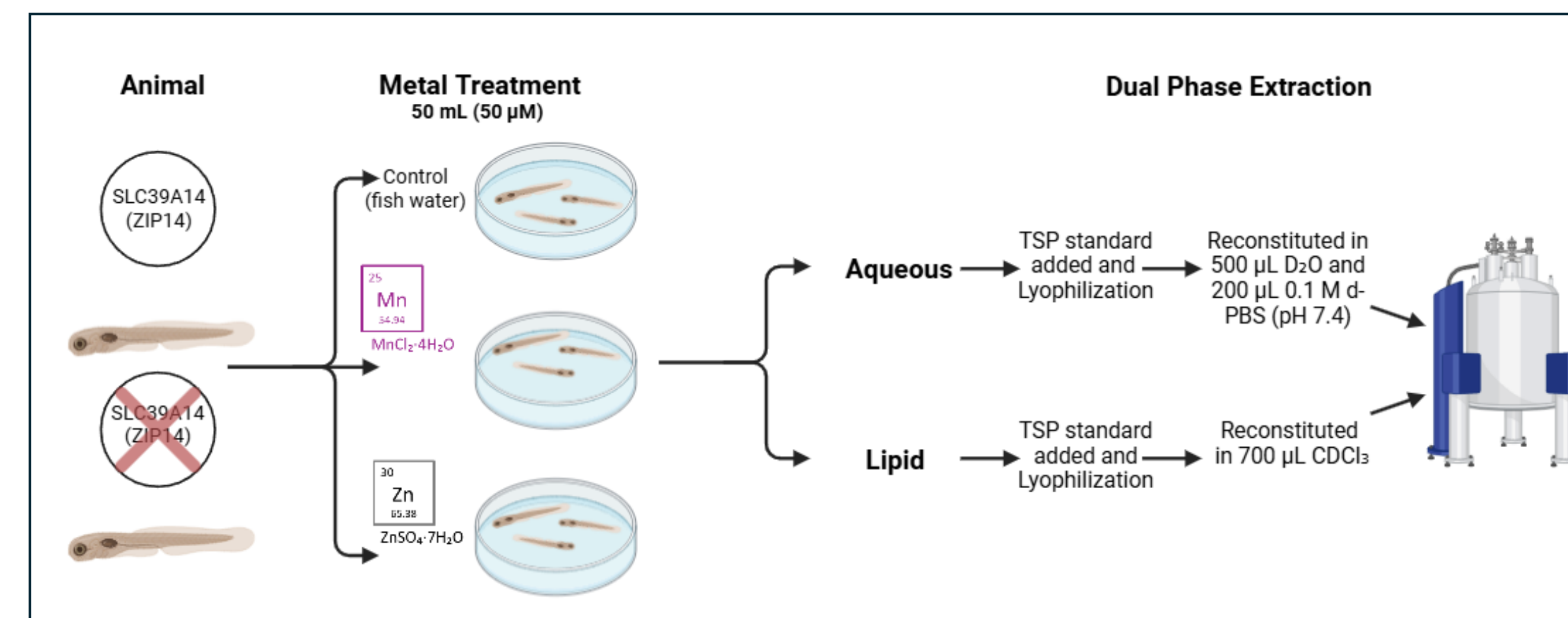


Figure 1. ZIP14 WT and ZIP14 KO zebrafish embryos treated with manganese, zinc, or control (no treatment, fish water only) followed by dual phase extraction and ¹H nuclear magnetic resonance (NMR) spectroscopy. Abbreviations: solute carrier family 39 member 14 (slc39a14), trimethylsilyl-[2,2,3,3-²H₄]-propionate (TSP), deuterium oxide (D₂O), deuteriochloroform (CDCl₃), deuterated phosphate buffer (d-PBS). Created using BioRender.

RESULTS

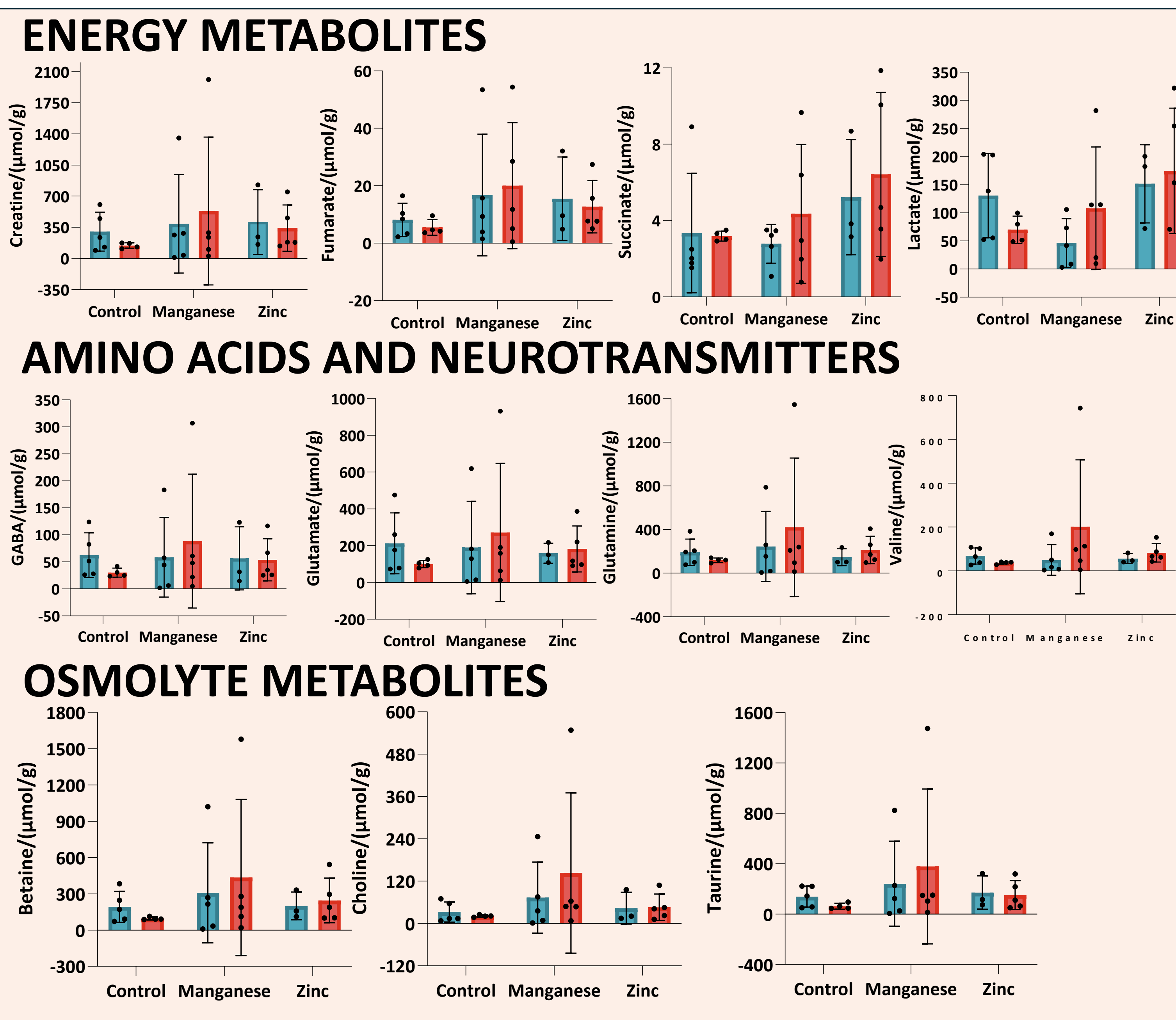


Figure 2. Normalised mean aqueous metabolite concentrations (µmol/g) obtained from dual phase extraction of zebrafish embryos from **wild-type ZIP14 (WT)** and **knock-out ZIP14 (KO)** samples. Normalisation was based on wet weights of zebrafish embryos. Values are the mean ± SD (µmol/g wet tissue). Creatinine, Glycine, Tyrosine were not shown.

- Concentrations of creatine, fumarate, succinate, lactate show no changes (confirmed via a 2-way ANOVA)
- Amino Acid and Neurotransmitter concentrations of GABA, glutamate, glutamine, and valine remain unchanged
- Osmolyte metabolite concentrations of betaine, choline, taurine remain unchanged
- However, minor trends were observed between metabolites in Figure 2 inferring metabolic disruptions seen in neurodegenerative disease

CONCLUSIONS

- ZIP14 loss produced only **non-significant metabolic trends**, suggesting either subtle Mn or Zn-specific effects or compensatory pathways that mask clear outcomes.
- ZIP14 loss may have altered Mn but not Zn-linked metabolism, with only **subtle trends** in amino acids and glycolytic intermediates, highlighting limited statistical power.
- Trend findings** align with Mn's established role in mitochondrial and neurotransmitter metabolism, but **absence of significance** or **changes** seen in prior studies suggest either compensatory mechanisms or insufficient sensitivity of ¹H NMR at this stage.
- Future validation with larger cohorts, a different **treatment regimen** and **targeted organ and organelle specific** (mitochondrial) **assays** are essential to determine whether these trends reflect true Mn and Zn-driven disruptions or methodological limitations and **integration with metallomics** to resolve whether observed trends reflect **true Mn and Zn metabolic** disruption due to ZIP14 loss.

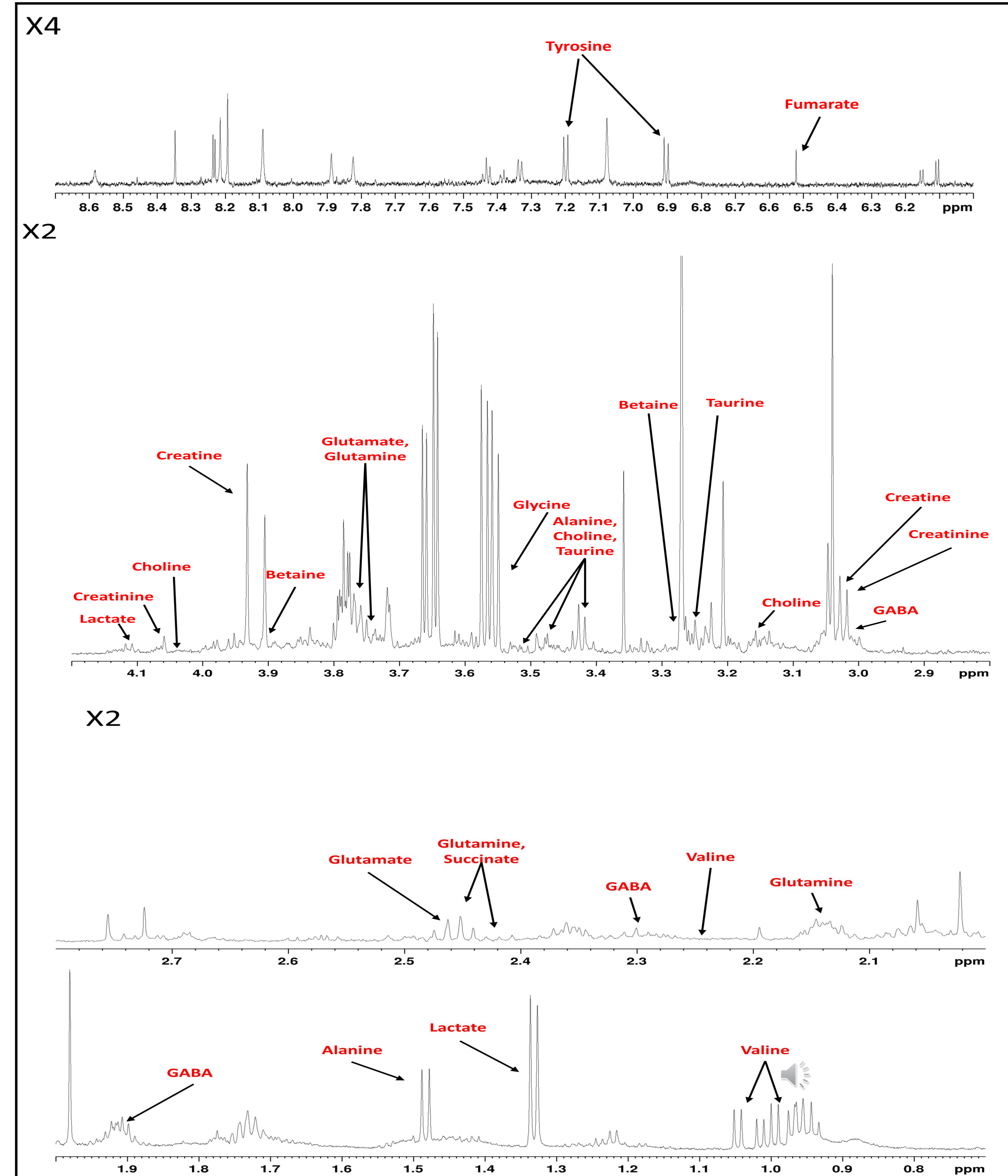


Figure 3. Partial ¹H nuclear magnetic resonance spectrum of the δ 0.7 – 8.7 ppm region (excluding δ 4.2 – 6.0 ppm) of the aqueous phase obtained following dual phase extraction with chloroform, methanol and water of control (fish water) treated ZIP14 KO zebrafish embryo. Assignments are based on Chenomx™, literature values. Abbreviation: γ-aminobutyrate (GABA). Gain multipliers (e.g., ×2, ×4) are indicated in the top left of the spectra region directly below spectrum and are relative to the full-scale spectrum.